WORKED SOLUTION

STEP II 2000, Question 14

Probability and Statistics • Distribution of the Median

STEP II 2000, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q14

This question has a full guided treatment in *Beyond STEP*, Section 13.4, *Mean or Median?*: https://beyondstep.keithmcnulty.org/ch13/mean-or-median.html

The key idea

The density of the median is a polynomial of the form $y^r(1-y)^s$, so the given integral does all the work: once for the normalising constant, then again with the power of y raised by one and by two for the first two moments. For the final part, both Y and $\bar{X}$ are centred at $\frac{1}{2}$, and with normal distributions the probability of landing within a fixed distance of the centre depends only on the standard deviation. The variable with the smaller spread wins.

Solution

The constant k

A density must integrate to 1. Using the given result with $r = s = n$,

$$1 = \int_0^1 k y^n (1-y)^n \, \mathrm{d}y = k \, \frac{n!\, n!}{(2n+1)!}, \qquad \text{so} \qquad k = \frac{(2n+1)!}{(n!)^2},$$

as required.

The mean

The density is symmetric about $y = \frac{1}{2}$ (replacing y by $1-y$ leaves it unchanged), so $\mathrm{E}(Y) = \frac{1}{2}$. To confirm this with the integral, take $r = n+1$, $s = n$:

$$\mathrm{E}(Y) = k \int_0^1 y^{n+1}(1-y)^n \, \mathrm{d}y = \frac{(2n+1)!}{(n!)^2} \cdot \frac{(n+1)!\, n!}{(2n+2)!} = \frac{n+1}{2n+2} = \frac{1}{2}.$$

The variance

Now take $r = n+2$, $s = n$:

$$\mathrm{E}(Y^2) = \frac{(2n+1)!}{(n!)^2} \cdot \frac{(n+2)!\,n!}{(2n+3)!} = \frac{(n+2)(n+1)}{(2n+3)(2n+2)} = \frac{n+2}{2(2n+3)}.$$

Hence

$$\mathrm{Var}(Y) = \frac{n+2}{2(2n+3)} - \frac{1}{4} = \frac{2(n+2)-(2n+3)}{4(2n+3)} = \frac{1}{4(2n+3)}.$$

Comparing the median with the mean

Each X_i is uniform on $[0,1]$, so $\mathrm{E}(X_i) = \frac{1}{2}$ and $\mathrm{Var}(X_i) = \frac{1}{12}$. Since the X_i are independent,

$$\mathrm{E}(\bar{X}) = \tfrac{1}{2}, \qquad \mathrm{Var}(\bar{X}) = \frac{1}{(2n+1)^2} \cdot (2n+1) \cdot \frac{1}{12} = \frac{1}{12(2n+1)}.$$

> **Ask yourself** Both variables are centred at $\frac{1}{2}$. What single number decides which is more likely to fall within $d/\sqrt{n}$ of the centre?

The standard deviation. For large n we may take $Y \sim \mathrm{N}(\frac{1}{2}, \sigma_Y^2)$ and $\bar{X} \sim \mathrm{N}(\frac{1}{2}, \sigma_X^2)$. If $W \sim \mathrm{N}(\frac{1}{2}, \sigma^2)$ and $c > 0$, then, writing Φ for the standard normal distribution function,

$$\mathrm{P}\big(|W - \tfrac{1}{2}| < c\big) = \mathrm{P}\Big(|Z| < \frac{c}{\sigma}\Big) = 2\Phi\Big(\frac{c}{\sigma}\Big) - 1,$$

where $Z \sim \mathrm{N}(0,1)$. This is a strictly *decreasing* function of σ, because Φ is strictly increasing. So with $c = d/\sqrt{n}$ the required inequality holds exactly when $\sigma_Y > \sigma_X$, that is, when $\mathrm{Var}(Y) > \mathrm{Var}(\bar{X})$:

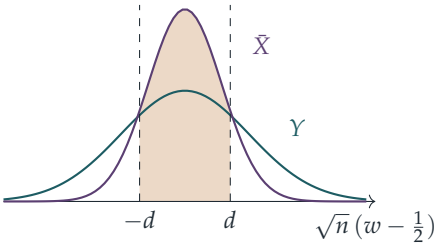
$$\begin{aligned} \frac{1}{4(2n+3)} > \frac{1}{12(2n+1)} &\iff 12(2n+1) > 4(2n+3) \\ &\iff 24n + 12 > 8n + 12 \\ &\iff n > 0. \end{aligned}$$

This is true for every $n \geq 1$. So once n is large enough for the normal approximations to be valid, the probability for Y is strictly less than the probability for $\bar{X}$, whatever the positive number d. This completes the proof.

To see how big the gap is, standardise:

$$\frac{d/\sqrt{n}}{\sigma_Y} = 2d\sqrt{\frac{2n+3}{n}} \to 2\sqrt{2}\,d, \qquad \frac{d/\sqrt{n}}{\sigma_X} = d\sqrt{\frac{12(2n+1)}{n}} \to 2\sqrt{6}\,d.$$

Asymptotically, the mean's standard deviation is smaller by a factor of $\sqrt{3}$.



The figure shows the two limiting densities on the scale $\sqrt{n}\,(w-\frac{1}{2})$. The narrower curve (the mean) puts more probability in any central band.

Checks

- $n=0$: the "median" of one uniform variable is just X_1, and our formulas give $k=1$ and $\mathrm{Var}(Y)=\frac{1}{12}$, the uniform variance, as they should.
- Both variances equal $\frac{1}{12}$ when $n=0$ and split apart for $n\geq 1$, which matches the inequality $n>0$ above.
- A simulation with $n=50$ (101 uniforms) gave $\mathrm{Var}(Y)\approx 0.00243$ against $1/412\approx 0.00243$. With $d=0.3$ it gave $\mathrm{P}(|Y-\frac{1}{2}|<d/\sqrt{n})\approx 0.607$ and $\mathrm{P}(|\bar{X}-\frac{1}{2}|<d/\sqrt{n})\approx 0.859$, close to the normal values 0.611 and 0.860.

Common slips

- Writing $\mathrm{Var}(\bar{X})=\frac{1}{12}$, or dividing by $2n+1$ only once instead of getting $(2n+1)^2$ in the denominator and then multiplying by $2n+1$.
- Getting the direction wrong: a *larger* variance gives a *smaller* probability of being near the centre.
- Forgetting that $\mathrm{E}(Y^2)$ uses $r=n+2$, not $r=2n$.

Beyond STEP

The density of the median is a Beta distribution, $\mathrm{Beta}(n+1,n+1)$, and the given integral is the Beta function. The comparison you have just made is the idea of *asymptotic relative efficiency*: for uniform data the sample mean estimates the centre with a third of the median's variance, while for heavy-tailed data the median can win. Order statistics and the Beta function are covered well in Blitzstein and Hwang, *Introduction to Probability*.

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